

“Short Leg Syndrome” and Heel Lift Therapy



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Session Objectives

- 1. Students will be able to understand why the name “Short Leg Syndrome” is used to date, what symptoms and/or findings comprise the syndrome, and understand true and apparent leg-length discrepancy. Students will be able to understand what anatomical landmarks are measured and considered with radiological postural evaluation
- 2. Students will be able to understand a "C-scoliotic response" and an "S-scoliotic response." Students will be able to understand and review the bony, ligamentous and muscular involvement in the lumbo-sacral-pelvic area, in relation to the “Short Leg Syndrome.”
- 3. Students will be able to understand how to organize the evaluation and examination for the "Short Leg Syndrome," including Special Tests. Students will be able to understand what heel lift therapy is, and to which side a heel lift would be added.

Session Objectives

- 4. Students will be able to understand how to utilize David Heilig's formula, for calculation of heel lift height based on the radiologically measured sacral base unleveling (SBU). Students will be able to consider differential diagnoses potentially contributing to the symptoms of “Short Leg Syndrome.”
- 5. Students will be able to understand the maximum height of heel lift that can be placed inside of a shoe, and how the heel lift height added to the outside of the shoe is managed. Students will be able to understand the goals of treatment for “Short Leg Syndrome.”

“Short Leg Syndrome”

- A misnomer, really. Name used due to historical precedent.
- Actual causes may be unrelated to leg lengths.
- A “syndrome” since there are multiple symptoms and findings.
 - Sacral base unleveling
 - Innominate rotation (compensatory)
 - Caudal curve side-bending away from side of low sacral base, most often.

“Short Leg Syndrome”

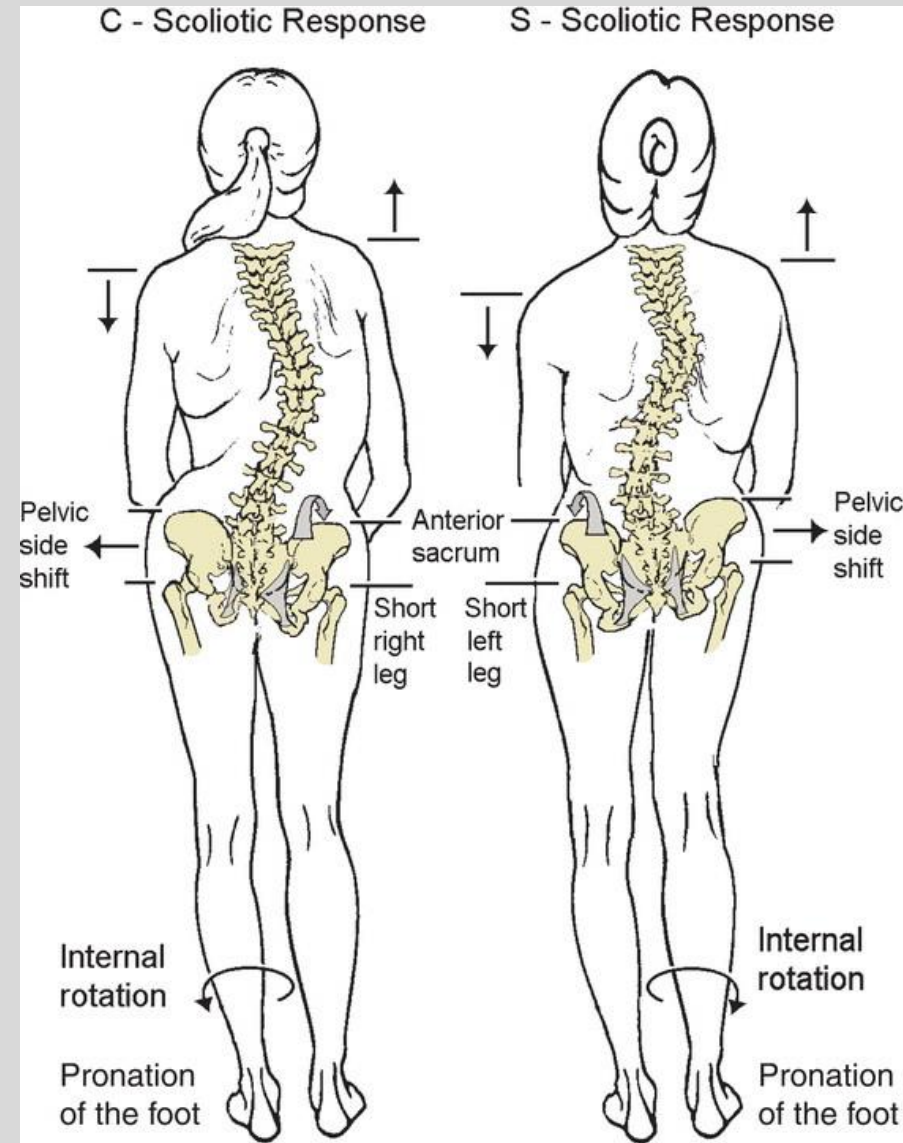
- Diagnosis and Treatment

- Treat all somatic dysfunction diagnoses as well as possible **before** sending patient for radiological evaluation.
- Radiology - Measure coronal values of iliac crest heights, femoral head heights, sacral base unleveling, compensatory scoliotic curves (type, location, degree).
- Remember: 2mm human error in measurement; 25% bony distortion.

“Short Leg Syndrome”

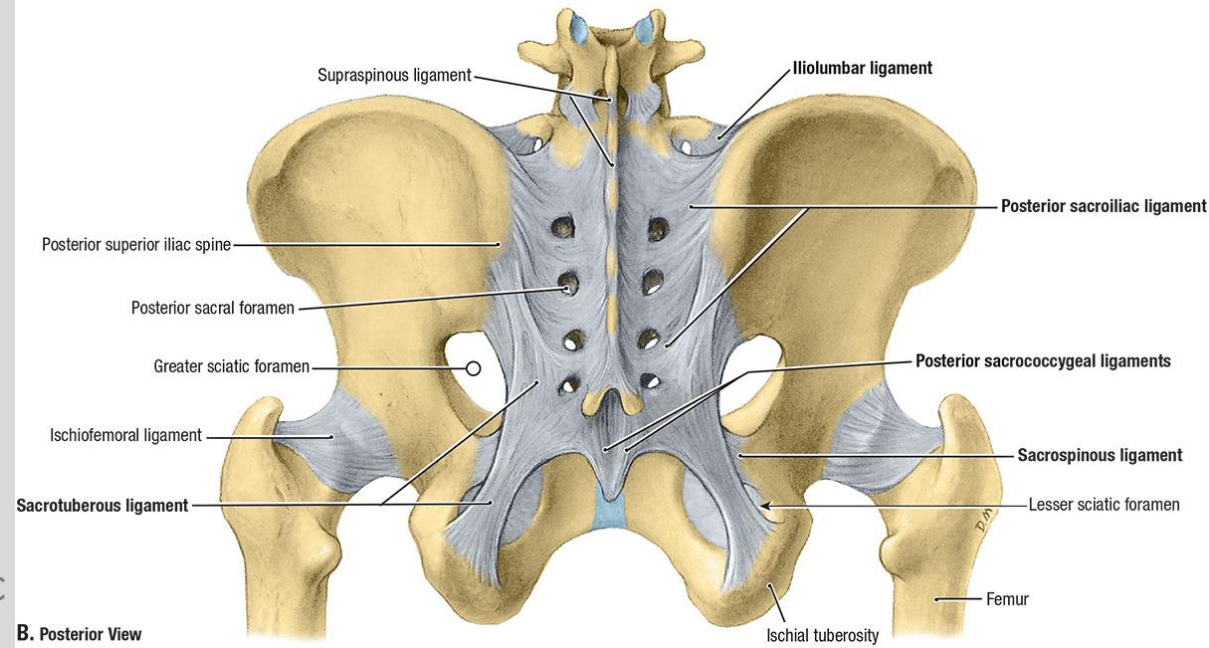
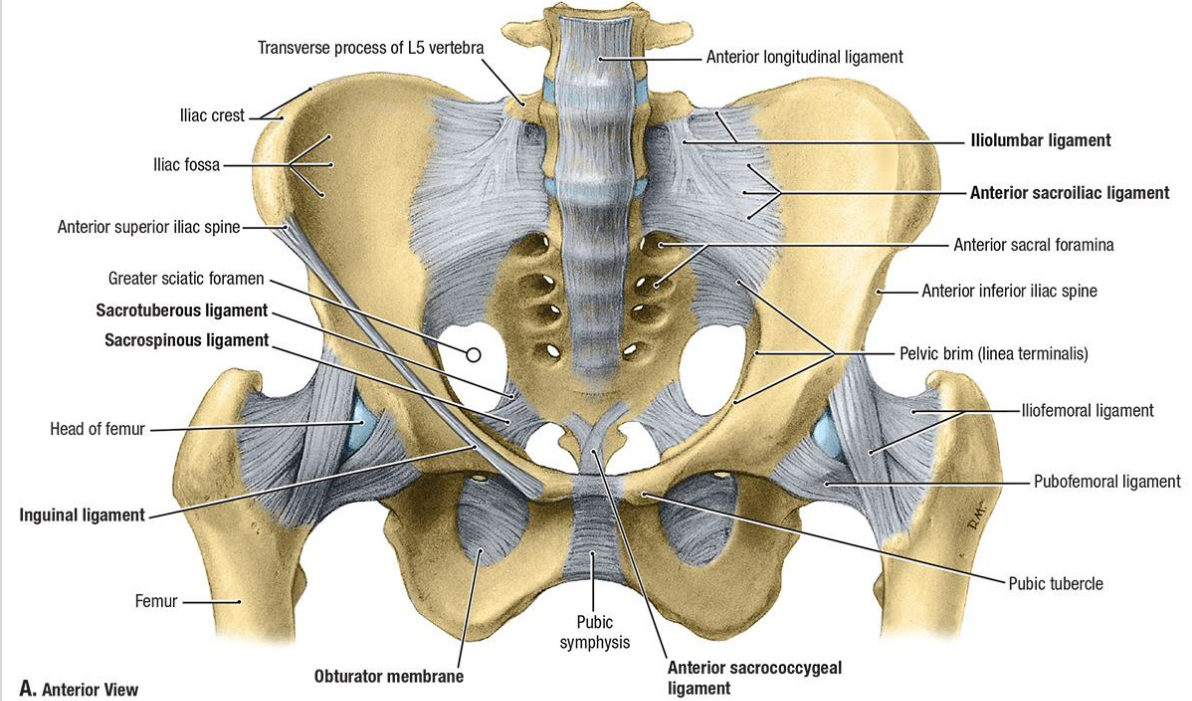
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Seffinger, Michael A., D.O., executive editor, **Foundations of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29: “Postural Considerations in Osteopathic Diagnosis and Treatment,”** see Heading of “Common Postural Clinical Conditions,” then see Figure 29.3, Typical Postural Compensation for Short Leg Syndrome, LINK: <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547478>

Ligaments

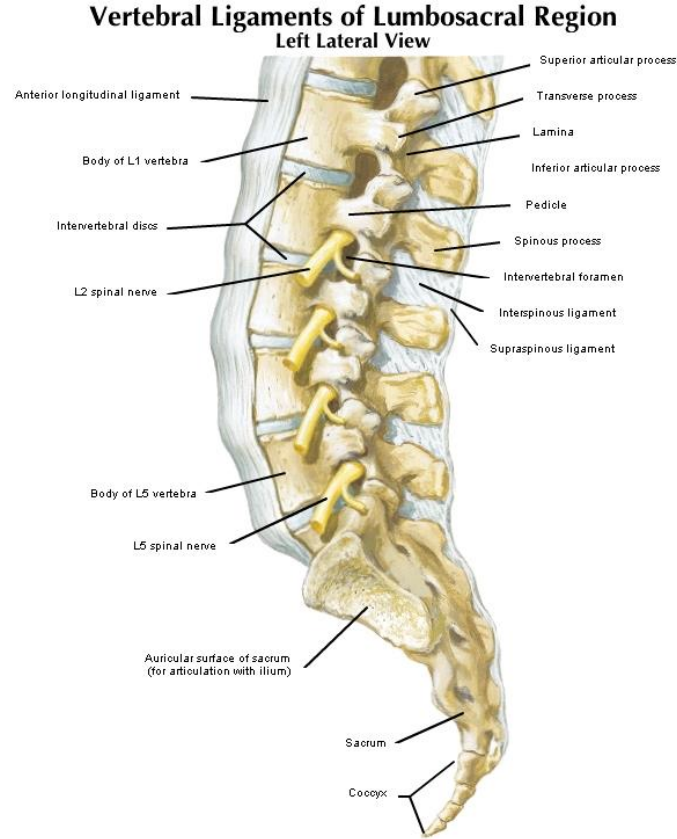


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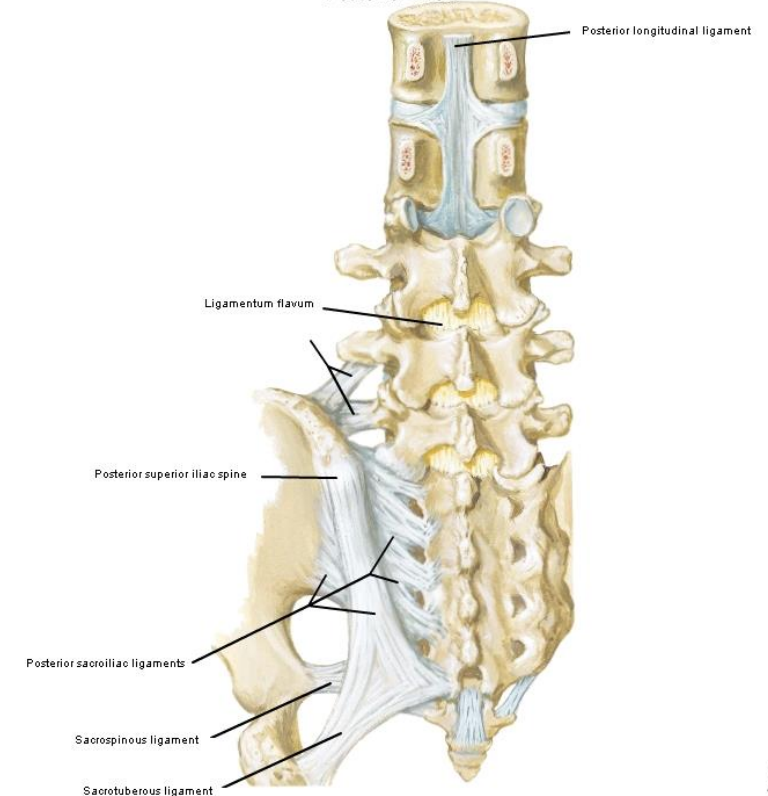
Ligaments

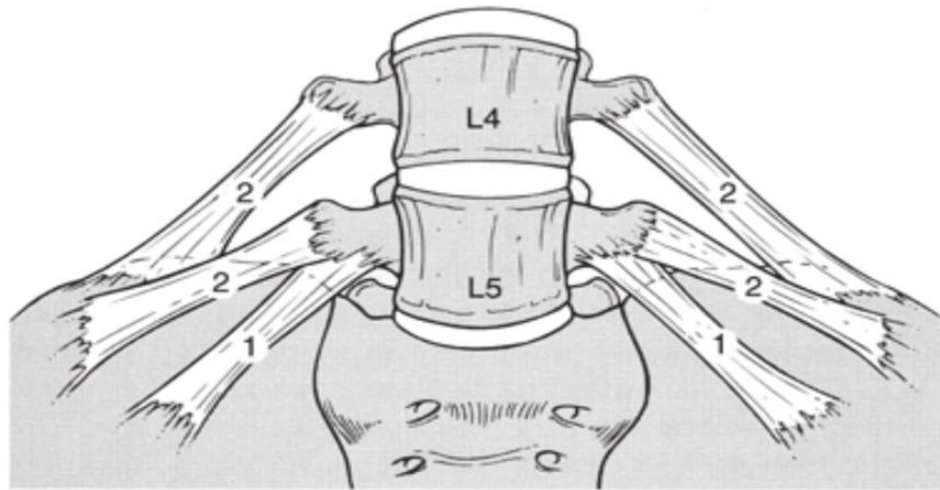
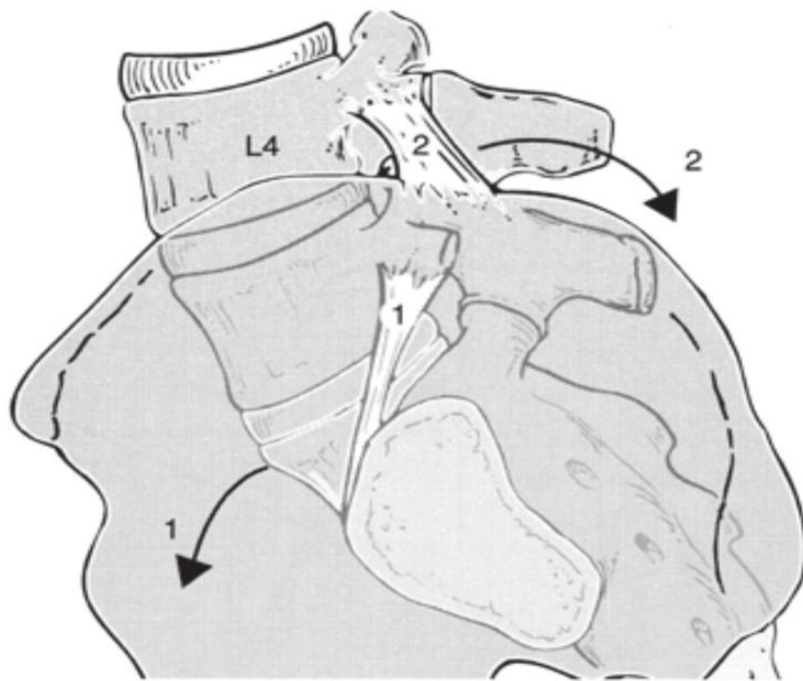
- Anterior Longitudinal Ligament
- Posterior longitudinal Ligament
- Ligamentum Flavum
- Interspinous Ligament
- Supraspinous Ligament
- Intertransverse Ligament
- Iliolumbar Ligament



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Vertebral Ligaments of Lumbosacral Region
Posterior View





A

B

Fig. 36.8 - From: Chapter 36: Postural Considerations in Osteopathic Diagnosis and Treatment, *Foundations of Osteopathic Medicine*, 3e, 2010, **A Anterior rotation of sacrum bilaterally stresses fibers labeled 1. Posterior rotation of innominates bilaterally stress fibers labeled 2. B Calcification of iliolumbar ligament is an excellent example of structural change resulting from excess functional demand.** (Radiograph from the Institute for Gravitational Strain Pathology Inc., with permission.); <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=50007348&bookid=757>

Iliolumbar ligament pain pattern from postural imbalance

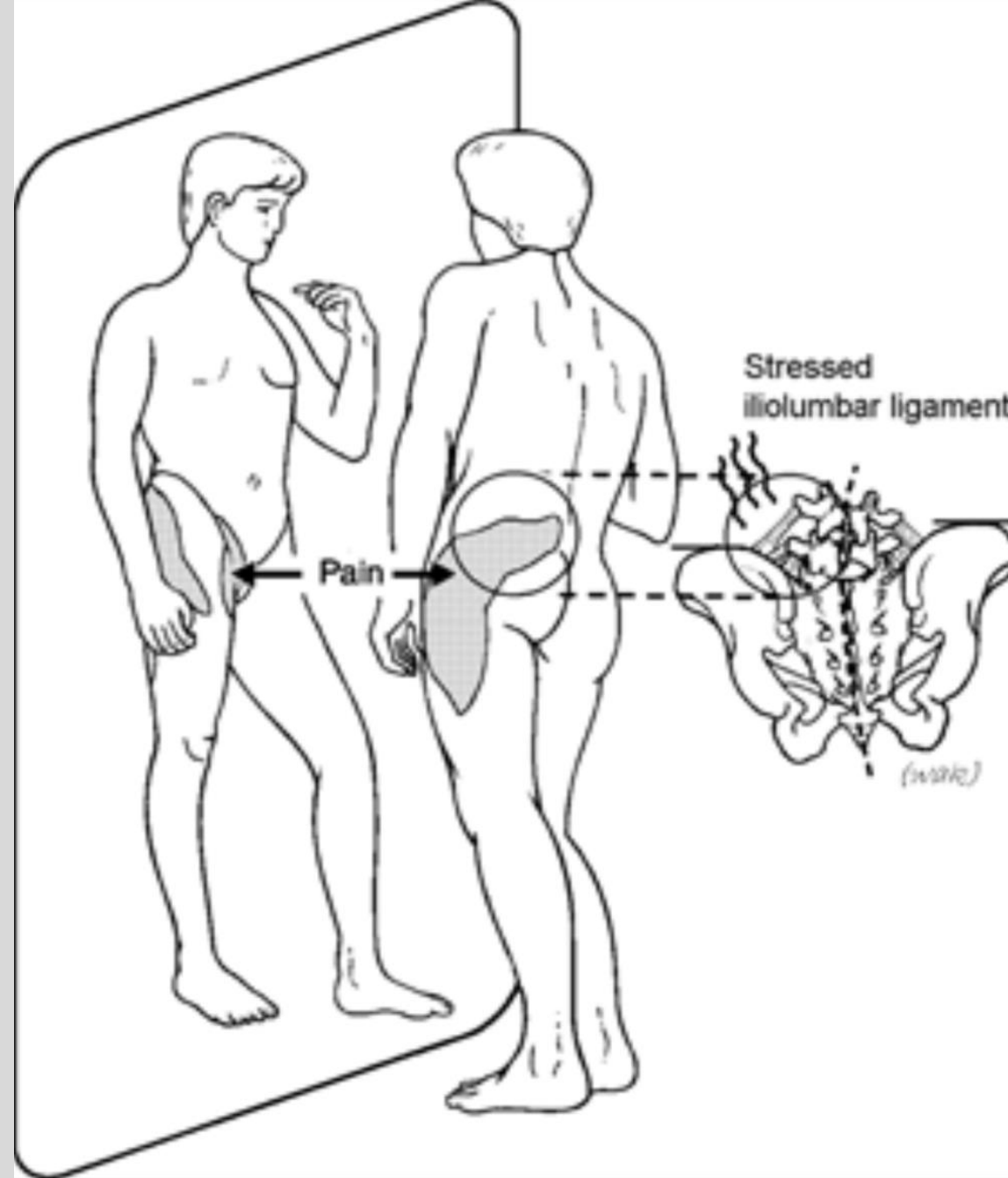
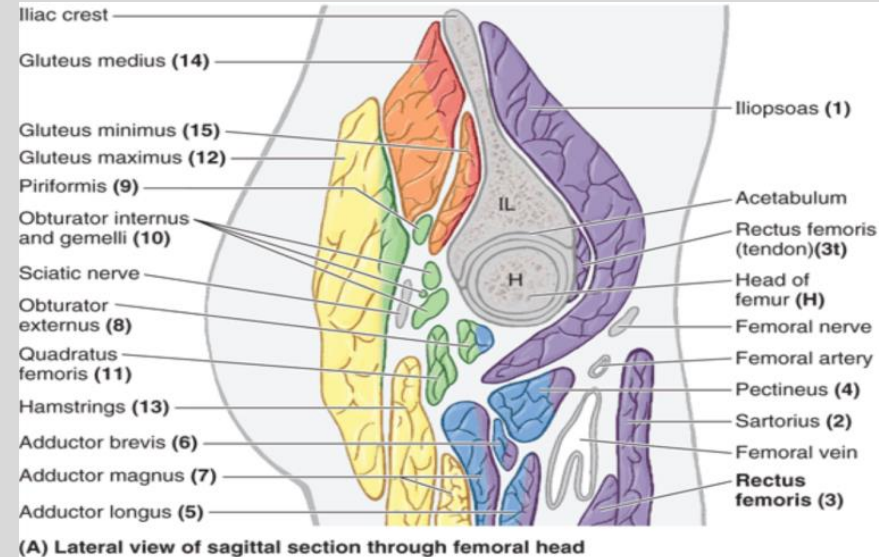
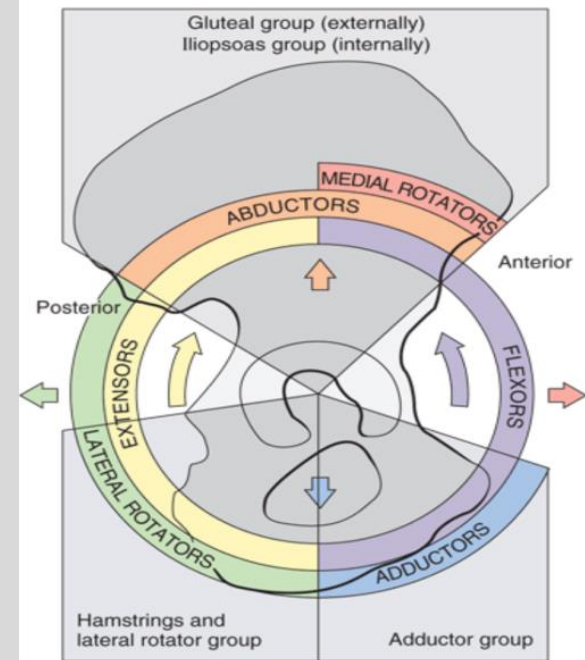


Fig. 36.9 - From: Chapter 36:
**Postural Considerations in
Osteopathic Diagnosis and
Treatment**, Foundations of
Osteopathic Medicine, 3e, 2010,
**Iliolumbar ligament pain pattern
from postural imbalance.**
(Courtesy of William Kuchera,
D.O., F.A.A.O.),
<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-jwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=50007348&bookid=757>

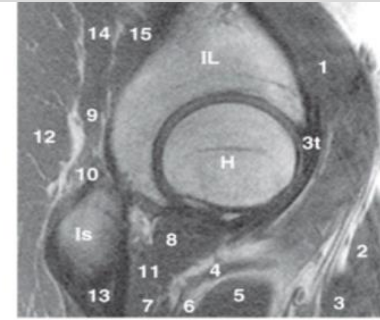
Muscles



(A) Lateral view of sagittal section through femoral head



(C) Diagrammatic lateral view



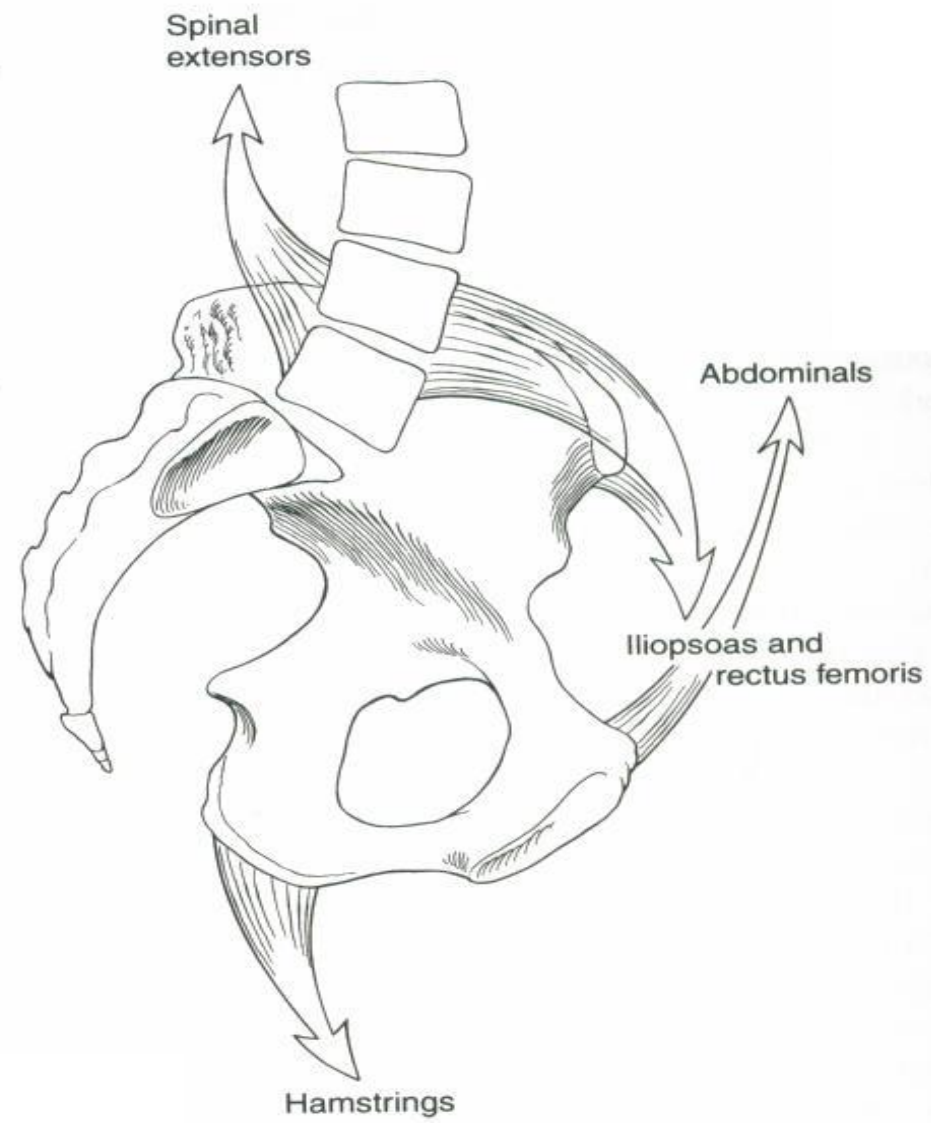
(B) MRI of hip joint, lateral view (Is, Ischium)

Functional groups of muscles acting at hip joint

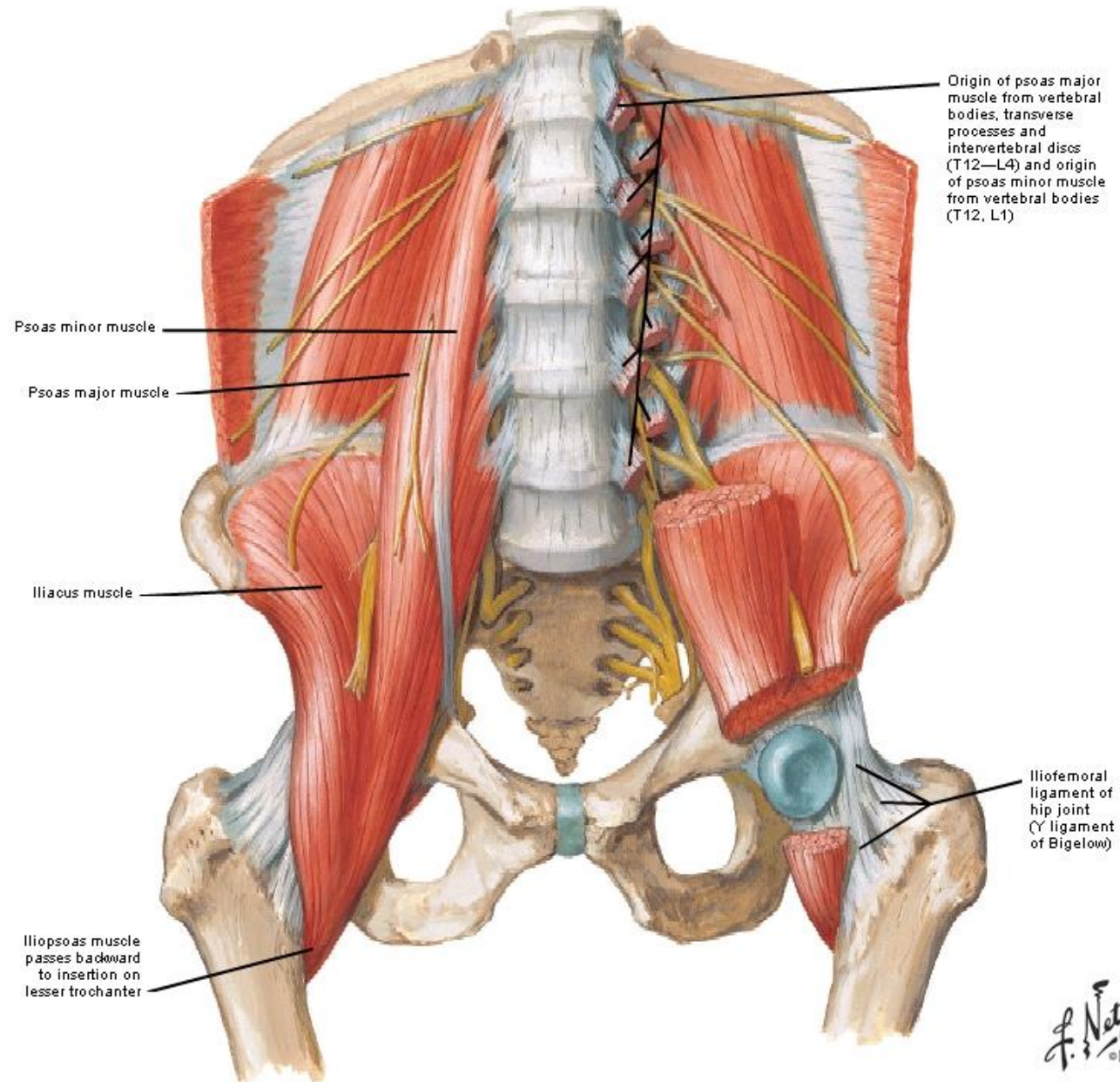
Flexors
Iliopsoas (1) Sartorius (2) Tensor fasciae latae Rectus femoris (3) tendon (3t) Pectineus (4) Adductor longus (5) Adductor brevis (6) Adductor magnus—anterior part (7) Gracilis
Adductors
Pectineus (4) Adductor longus (5) Adductor brevis (6) Adductor magnus (7) Obturator externus (8) Gracilis
Lateral rotators
Obturator externus (8) Piriformis (9) Obturator internus (10) Gemelli (10) Quadratus femoris (11) Gluteus maximus (12)
Extensors
Gluteus maximus (12) Hamstrings: (13) Semitendinosus Semimembranosus Long head, biceps femoris Adductor magnus—posterior part
Abductors
Gluteus medius (14) Gluteus minimus (15) Tensor fasciae latae
Medial rotators
Gluteus medius (14) Gluteus minimus (15) } Anterior parts Tensor fasciae latae

Fig. 5.83 - From: Lower Limb, Clinically Oriented Anatomy, 7e, 2013, Relations of hip joint and muscles producing movements of joint. A. Sagittal section of the hip joint showing the muscles, vessels, and nerves related to it. The muscles are color coded to indicate their function(s). Applying Hilton's law, it is possible to deduce the innervation of the hip joint by knowing which muscles directly cross and act on the joint and their nerve supply. B. The relative positions of the muscles producing movements of the hip joint and the direction of the movement are demonstrated.

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsscience.com.nyit.idm.oclc.org/book.aspx?bookid=739>



Psoas and Iliacus Muscles

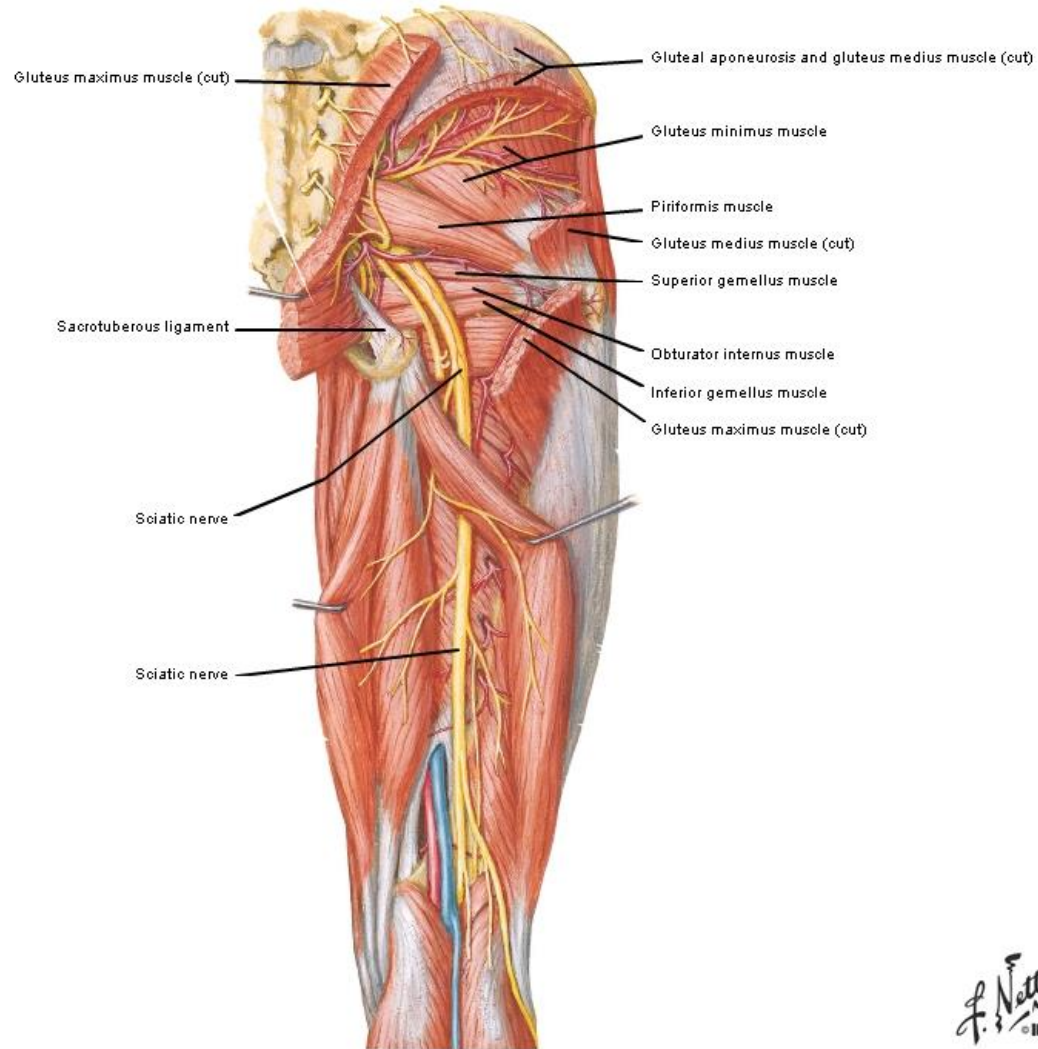


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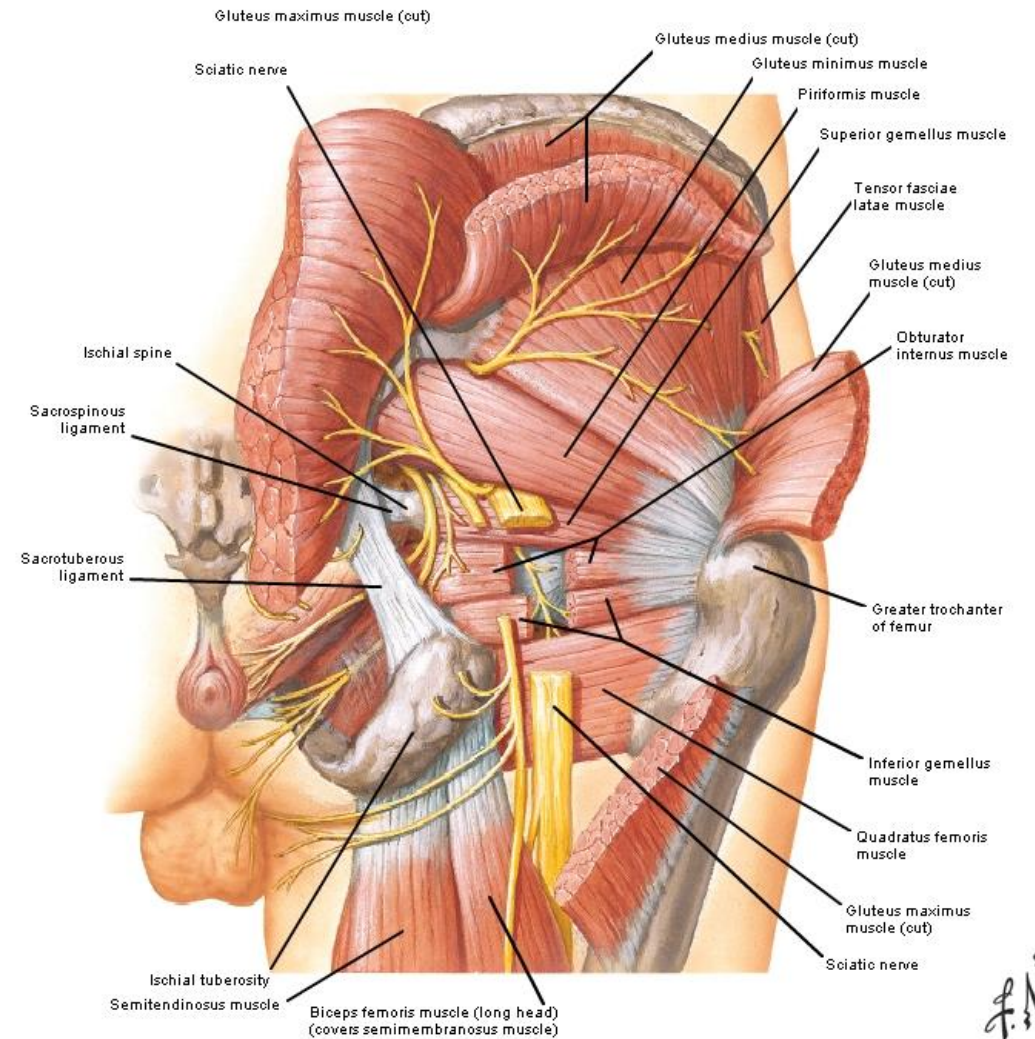
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Arteries and Nerves of Thigh (Deep Dissection) Posterior View



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Nerves of Hip and Buttock



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Evaluation

- Begins with H & P
- Acute
- Chronic
- Age
- Location of pain
- Timing of pain
- Review of medications

Examination

- Observation of stance
- Observation of gait
- Leg length discrepancy
- Range of motion- active and passive
- Muscle strength with resistance
- Neurologic examination
- Special tests
- Osteopathic Structural Examination

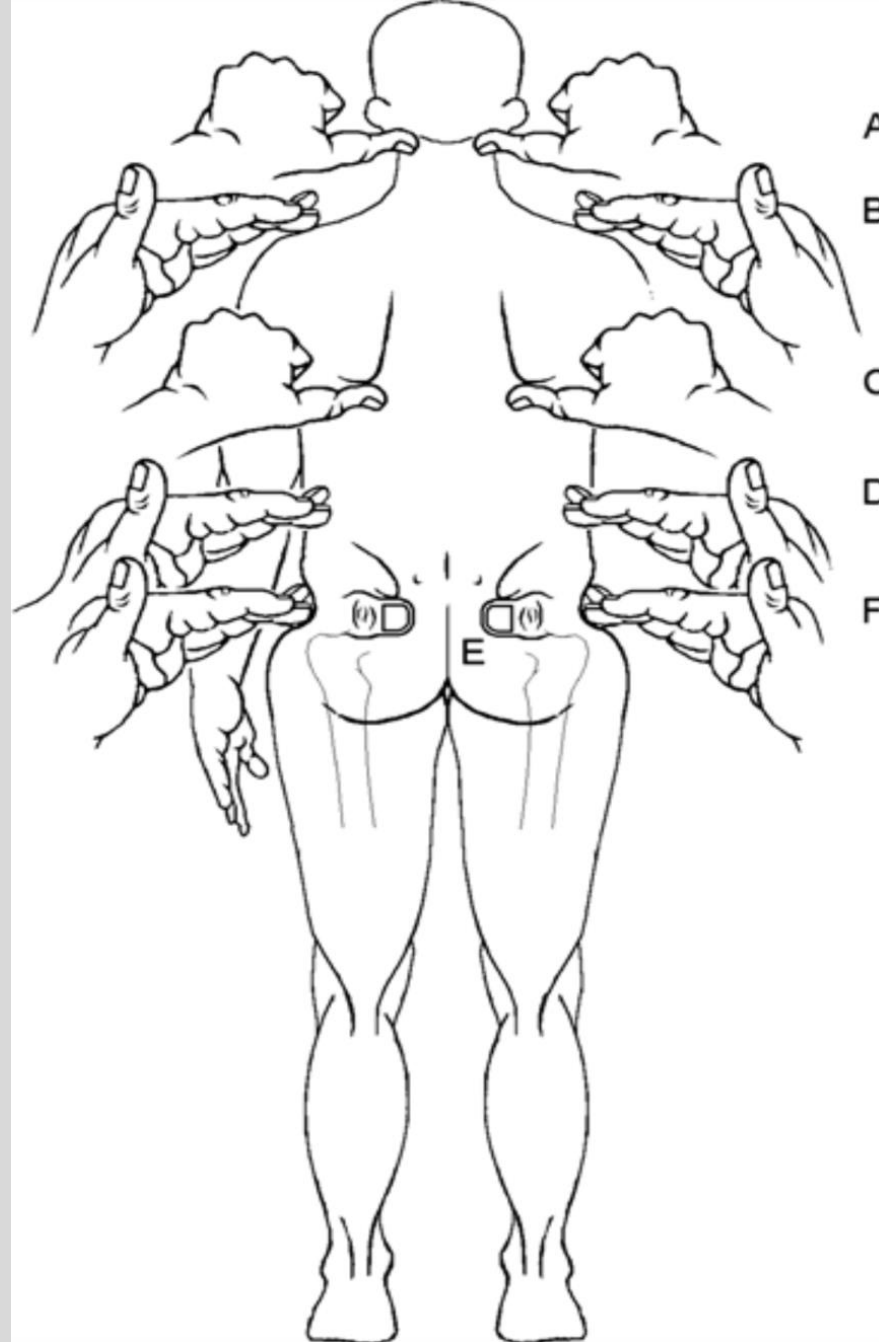
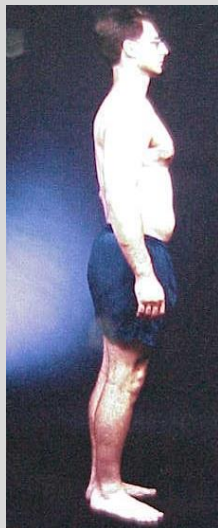


Fig. 36-12 - From: Chapter 36: **Postural Considerations in Osteopathic Diagnosis and Treatment**, Foundations of Osteopathic Medicine, 3e, 2010, **Levelness of horizontal planes**. A Occipital plane. B Shoulder plane. C Scapular plane. (inferior) D Iliac crest plane. E PSIS plane. F Greater trochanteric plane. (Courtesy of William Kuchera, D.O., F.A.A.O.), <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=50007348&bookid=757>

Observation of Stance

- Iliac crest heights level
- ASIS & PSIS level with contralateral side
- Increased lordosis of the lumbar spine
- Orientation of knees and feet



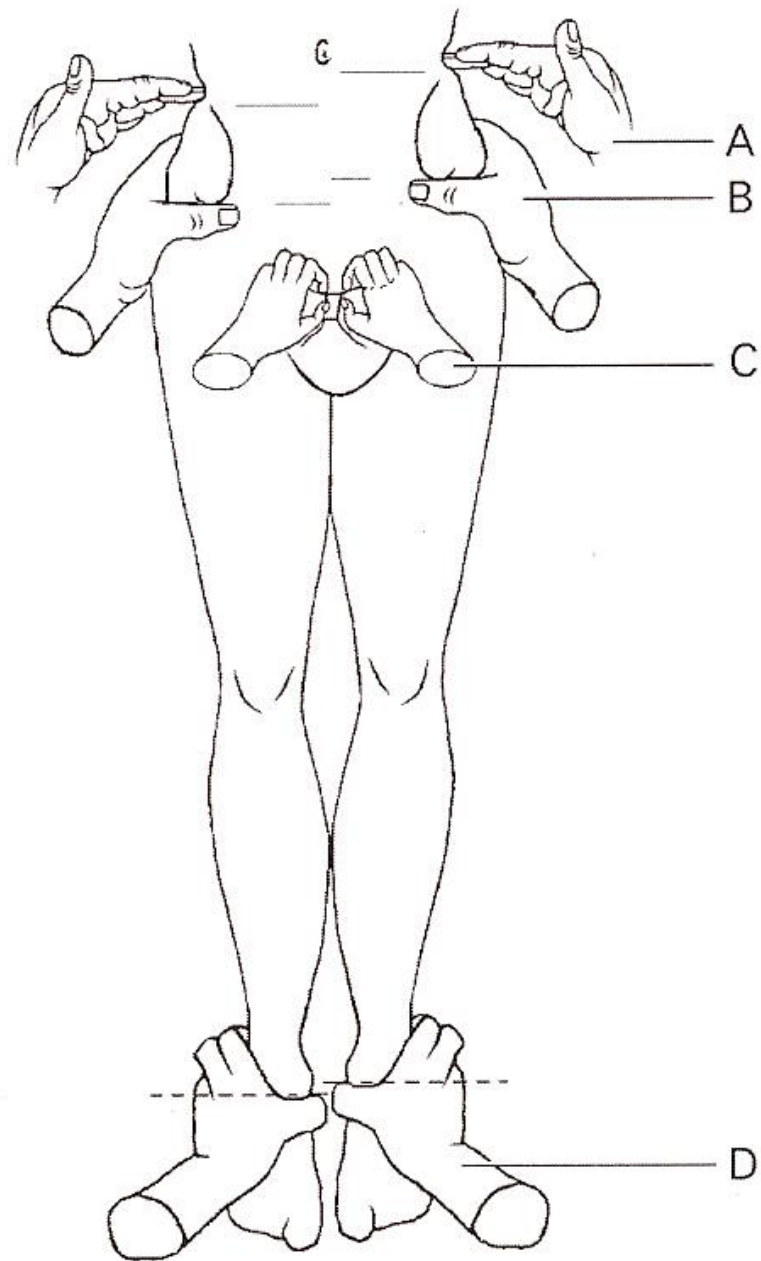


FIGURE 44.7. Anterior examination of bony landmarks (horizontal planes) of the body.

Leg Length Discrepancy

Apparent discrepancy- nonfixed pt to fixed pt

- Establish no true discrepancy exists first
- Check for pelvic torsion

True discrepancy-

Check fixed point to
fixed point

- ASIS to med. malleolus
- Femurs & tibias

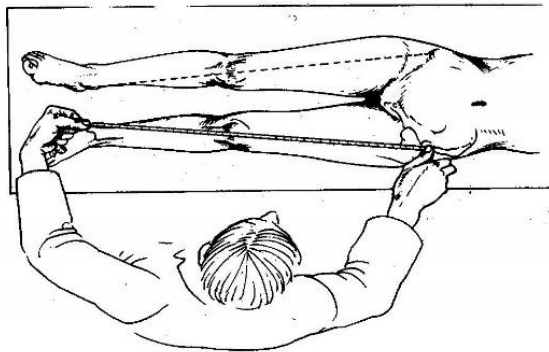


Fig. 57. Measure from one fixed bony point to another to find true leg length.

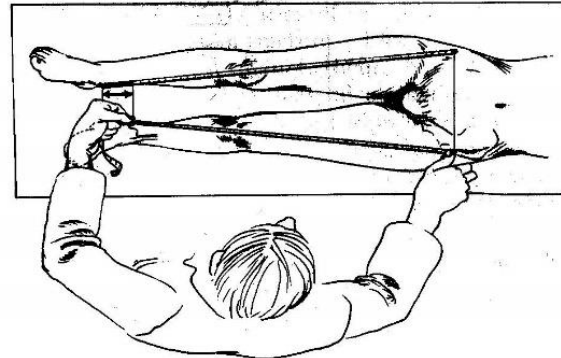


Fig. 58. True leg length discrepancy.

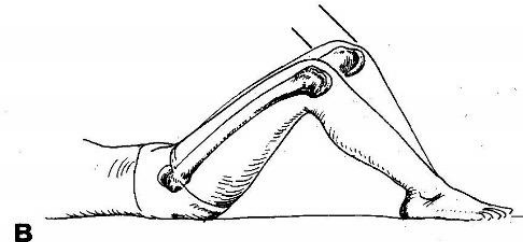
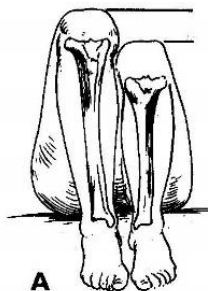


Fig. 59. A. Tibial length discrepancy. B. Femoral length discrepancy.

Special Tests

- Trendelenburg- gluteus medius weakness
- Straight leg raising- lumbar radiculopathy, contracted hamstring
- Stinchfield- intra-articular hip pathology
- FABERE/Patrick's Test- flexion, abduction, ext. rotation, extension. Hip, SI joint pathology
- Fadir/Fair- flexion, adduction, internal rotation. Piriformis, hip
- Thomas- iliopsoas contraction
- Erichsen's- SI & early spondylitic arthritis
- Ober's- contracture of ITB (iliotibial band)
- The (Adams) forward bend test- check for any rib hump(s)
- Hip Drop- assess lumbar involvement

Hip Drop Test

Lateral Lumbar Flexion: Hip Drop Test

Another method for assessing the ability of the lumbar spine to side-bend without a significant contribution from the thoracic spine is the *hip drop test*. The patient is standing with her weight equally distributed and the feet 4 to 6 in apart. The physician stands behind the patient with her eyes relatively level with the lumbar region and may kneel or squat to do so.



1. The patient is instructed to bend one knee, keeping both feet flat on the ground and the opposite knee straight (see **Figure**).
2. The patient should allow the compensatory shift in body weight distribution.
3. The physician notes the fluidity and amount of motion as the lumbar spine curves laterally to compensate for the drop in the pelvis on the side of the bent knee. The amount of drop in the iliac crest is noted.
4. The process is repeated to the contralateral side.
5. The right and left side-bending of the lumbar spine are compared. The iliac crest that drops the greatest amount is the side opposite that to which the lumbar spine side-bends the most, or, stated another way, the side of greatest drop is the side of restricted side-bending. The spinal curve should be smooth and unimpeded.

DiGiovanna, EL, Amen, CJ, and Burns, DK, An Osteopathic Approach to Diagnosis and Treatment, Fourth edition, Wolters Kluwer, 2021, Chapter 47 – “Evaluation of the Lumbar Spine,” by Eileen DiGiovanna, under the “Physical Examination” Heading, scroll down to “Lateral Lumbar Flexion: Hip Drop Test,” LINK: <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248813509&bookid=2969#248813514>

Hip Drop Test

https://nyitmedicine.mediaspace.kaltura.com/media/Hip%20Drop%20Test/1_4xh5hcs9

- Pt. is instructed to bend one knee, while keeping both feet flat on the ground (opposite knee is straight).
- Physician should take note of fluidity and amount of motion as lumbar spine curves laterally (compensating for drop in the pelvis on the BENT KNEE side).
- Note the amount of drop of the iliac crest.
- Repeat on the contralateral side.
- Compare the right and left side bending of the lumbar spine.
- The iliac crest that drops more is the side of restricted sidebending.



Adams Forward Bend Test



- Pt. stands in front of physician with feet together.
- Ask pt. to round her/his head forward & bend forward.
- Physician is looking to see if there is a difference between the right and left side in the area of the rib cage (looking for a rib hump).
- Sample findings: in a pt. who is SB left & rotated right → right rib cage would be more posterior compared to the left side.
- Then, ask patient to side bend towards the side of the rib hump (convexity). If the rib hump disappears, the curve is functional, if it remains, the curve is more structural in nature.

Sacral Base Unleveling, X-ray view

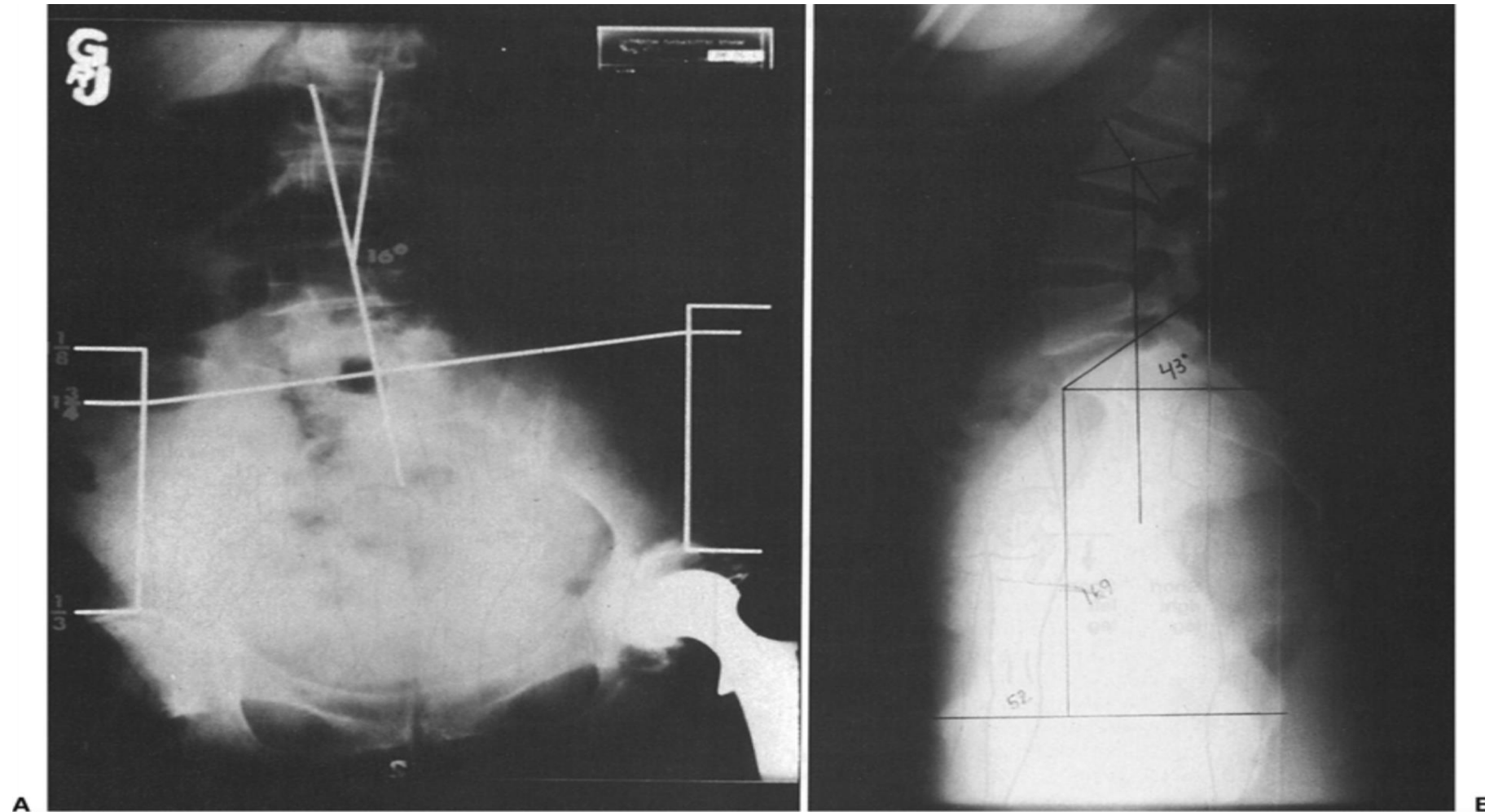


Fig. 38-6 - From: Chapter 36: Postural Considerations in Osteopathic Diagnosis and Treatment, Foundations of Osteopathic Medicine e, 3e, 2010, X-ray image marked for coronal (A) and horizontal (B) plane measurements. Ideal values would be no side-to-side unleveling of any of these landmarks and no pelvic rotation, <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=50007348&bookid=757>

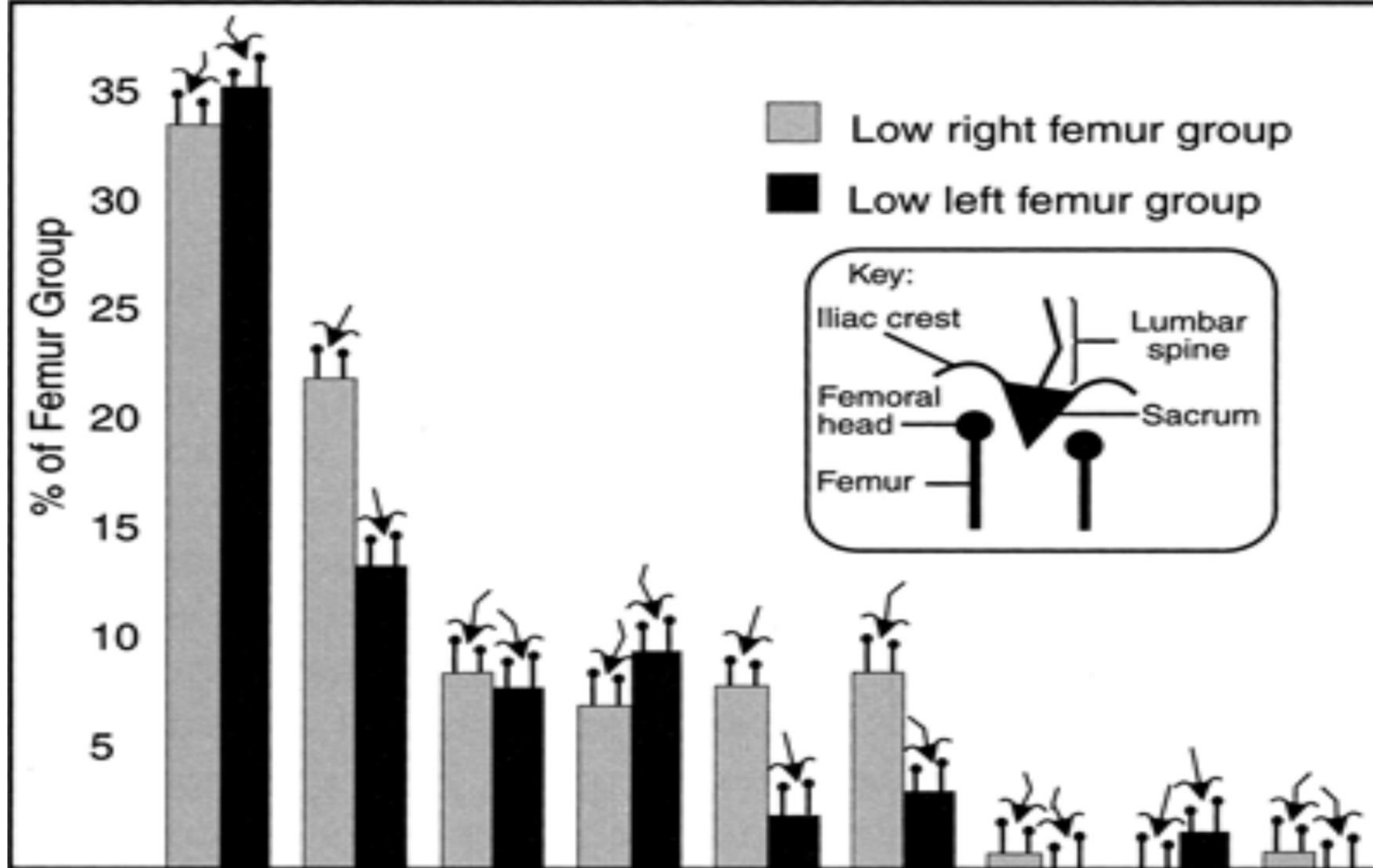
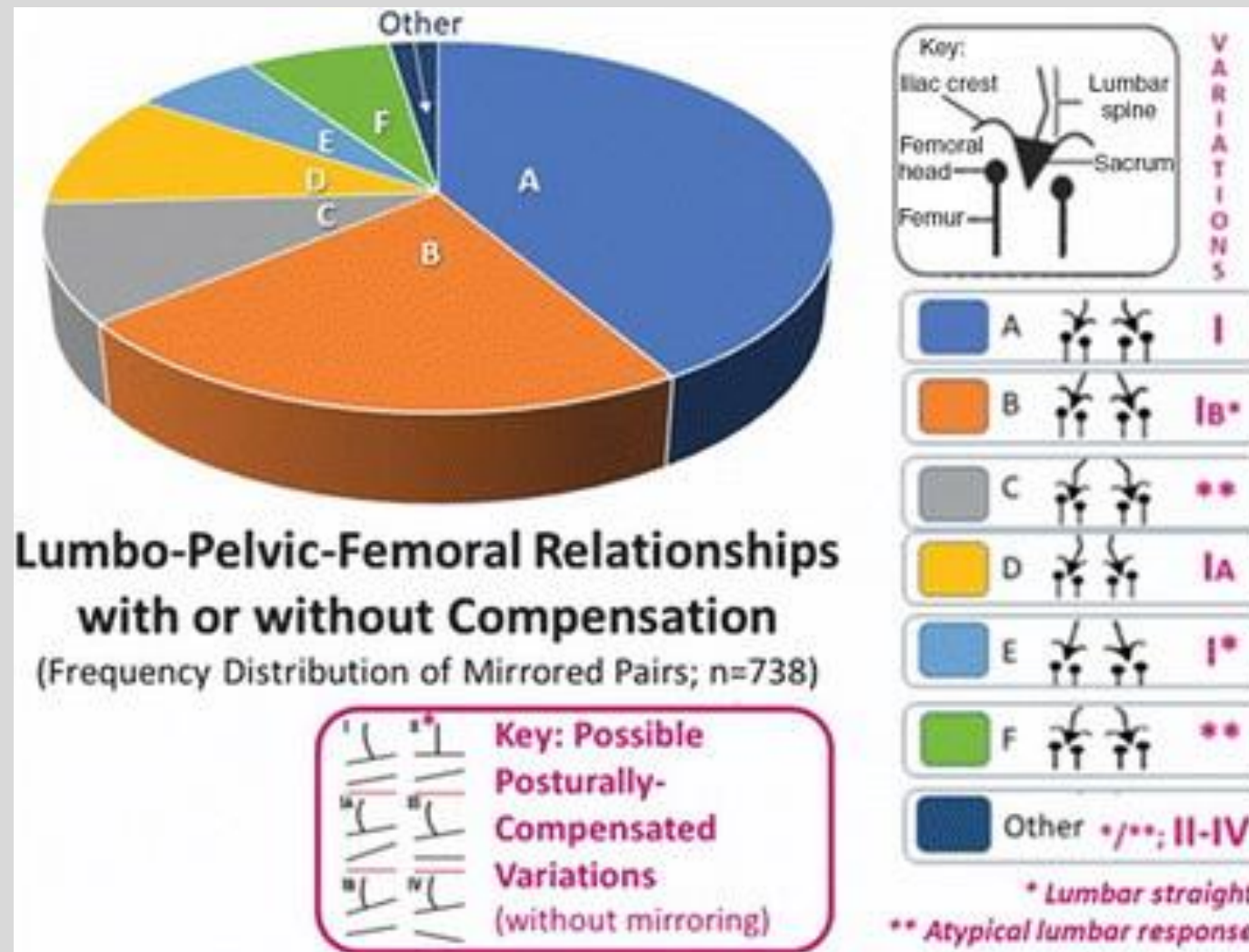


Fig. 36-28 - From: Chapter 36: Postural Considerations in Osteopathic Diagnosis and Treatment, Foundations of Osteopathic Medicine, 3e, 2010, Frequency distribution of lumbosacral relationships to unlevel femoral head heights. N = 738. (From Kirksville College of Osteopathic Medicine, Kirksville, MO, with permission.), <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=50007348&bookid=757>



Legend: Frequency distribution of possible lumbo-pelvic-femoral relationships to unlevel femoral head heights.

“Short Leg Syndrome”

- Heel Lift Therapy
 - OMT is done first.
 - Heel lift is added to the heel on the side of the lowered sacral base.
 - Acknowledge the postural realignment and reeducation that the patient will experience in the whole body.
 - “Heel Lift treatment using the **osteopathic postural model** is always combined with OMT”



Cork Heel Lift



Cork Heel Lift with Leather on top



Rubber Heel Lifts



Clearly Adjustable Heel Lift



Blue Foam Heel Lift
(needs to be firm)

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L \leq [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

“Short Leg Syndrome”

Case 1: A patient has healed from a right femoral fracture, right femur is now longer than the left femur, and he has 1/2 inch sacral base unleveling (SBU). The new asymmetry is of minimal duration, and the patient has no compensatory changes. Using the Heilig Formula, how much heel lift would you provide, and on which side?

- A 1/2 inch, right side
- B 1/2 inch, left side
- C 3/4 inch, right side
- D 3/4 inch, left side
- E 7/8 inch, right side
- F 7/8 inch, left side

Sample Heilig Formula Calculations

- Case 1: Following fracture, minimal duration, no compensatory changes
- $L = [SBU]/[D+C] = \frac{1}{2}''/[1 + 0] = \frac{1}{2}''$

“Short Leg Syndrome”

Case 1: A patient has healed from a right femoral fracture, right femur is now longer than the left femur, and he has 1/2 inch sacral base unleveling (SBU). The new asymmetry is of minimal duration, and the patient has no compensatory changes. Using the Heilig Formula, how much heel lift would you provide, and on which side?

- A 1/2 inch, right side
- ***B 1/2 inch, left side**
- C 3/4 inch, right side
- D 3/4 inch, left side
- E 7/8 inch, right side
- F 7/8 inch, left side

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L < \text{or } = \text{ to } [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

“Short Leg Syndrome”

Case 2: A patient is 35 years old, and had a right lower extremity injury in early youth at age 10 which led to his 1/2 inch right lower sacral base unleveling (SBU). He has minimal compensation with side bending and rotation changes noted. Using the Heilig Formula, how much of a heel lift, and on which side, would you prescribe for this patient?

- A 1/8 inch, right side
- B 1/8 inch, left side
- C 1/6 inch, right side
- D 1/6 inch, left side
- E 1/4 inch, right side
- F 1/4 inch, left side

Sample Heilig Formula Calculations

- Case 2: Patient age 35, injured in early youth at age 10, minimal compensation with side bending and rotation changes noted
- $L = [SBU]/[D+C] = \frac{1}{2}''/[2 + 1] = \frac{1}{2}''/3 = 1/6''$

“Short Leg Syndrome”

Case 2: A patient is 35 years old, and had a right lower extremity injury in early youth at age 10 which led to his 1/2 inch right lower sacral base unleveling (SBU). He has minimal compensation with side bending and rotation changes noted. Using the Heilig Formula, how much of a heel lift, and on which side, would you prescribe for this patient?

- A 1/8 inch, right side
- B 1/8 inch, left side
- ***C 1/6 inch, right side**
- D 1/6 inch, left side
- E 1/4 inch, right side
- F 1/4 inch, left side

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L \leq [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

“Short Leg Syndrome”

Case 3: A 75-year-old patient had an injury in his youth at age 15 that led to his current 1/2 inch left lower sacral base unleveling (SBU). He has lumbar vertebral endplates with horizontal growths and spurring, marked rotation. How much of a heel lift would you prescribe, and on which side?

- A 7/8 inch, left side
- B 7/8 inch, right side
- C 1/10 inch, left side
- D 1/10 inch, right side
- E 1/4 inch, left side
- F 1/4 inch, right side

Sample Heilig Formula Calculations

- Case 3: Patient age 75, injured in youth at age 15, spurring, horizontal endplate development, rotation is marked
- $L = [SBU]/[D+C] = \frac{1}{2}''/[3 + 2] = \frac{1}{2}''/5 = 1/10''$

“Short Leg Syndrome”

Case 3: A 75-year-old patient had an injury in his youth at age 15 that led to his current 1/2 inch left lower sacral base unleveling (SBU). He has lumbar vertebral endplates with horizontal growths and spurring, marked rotation. How much of a heel lift would you prescribe, and on which side?

- A 7/8 inch, left side
- B 7/8 inch, right side
- ***C 1/10 inch, left side**
- D 1/10 inch, right side
- E 1/4 inch, left side
- F 1/4 inch, right side

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L \leq [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

“Short Leg Syndrome”

Case 4: A 26-year-old patient had polio, which affected her right leg, in her youth at age 12. She has 1/2 inch right lower sacral base unleveling, and no compensatory changes. How much of a heel lift would you prescribe, and on which side?

- A 1/2 inch, left side
- B 1/2 inch, right side
- C 1/4 inch, left side
- D 1/4 inch, right side
- E 3/4 inch, left side
- F 3/4 inch, right side

Sample Heilig Formula Calculations

- Case 4: Patient age 26, polio affecting right leg in youth at age 12, no compensatory change
- $L = [SBU]/[D+C] = \frac{1}{2}''/[2 + 0] = \frac{1}{2}''/2 = \frac{1}{4}''$

“Short Leg Syndrome”

Case 4: A 26-year-old patient had polio, which affected her right leg, in her youth at age 12. She has 1/2 inch right lower sacral base unleveling, and no compensatory changes. How much of a heel lift would you prescribe, and on which side?

- A 1/2 inch, left side
- B 1/2 inch, right side
- C 1/4 inch, left side
- ***D 1/4 inch, right side**
- E 3/4 inch, left side
- F 3/4 inch, right side

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L \leq [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

Diagnosis & Evaluation (cont.)

- Before adding a lift...
- Is short leg due to a pronated foot? Tx with orthotics (& ?lift)
- Is there a torsion or shear in the pelvis? Tx with OMT (& ?lift)
- Is short leg due to Short Leg Syndrome?

“Short Leg Syndrome”

- The standing flexion test should become negative, once a proper heel lift has been put in place, and the patient has no more pelvic or lower extremity dysfunctions.
- Maximum lift inside a shoe is $\frac{1}{4}$ inch.
- Maximum of $\frac{1}{2}$ inch between patient's heel and the floor. Variable placement of lift materials is possible.

“Short Leg Syndrome”

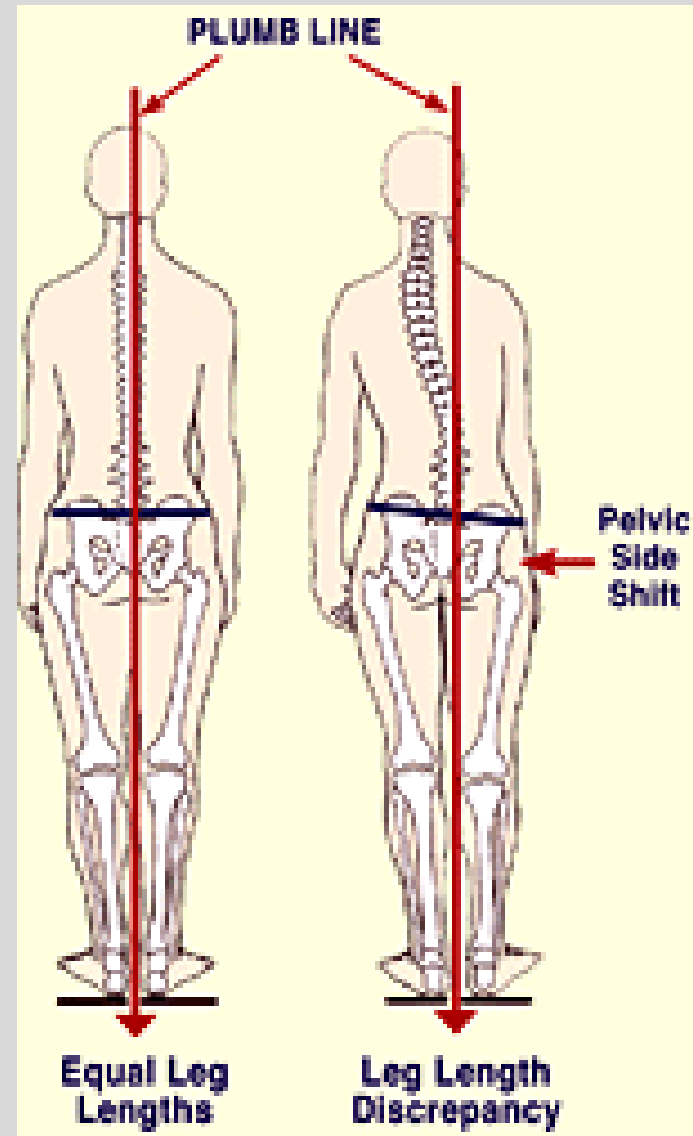
- Compressive forces can cause a bone to grow faster.
- Therefore, in a growing child, a lift under a short leg may cause that leg to grow faster.
- Important for the physician to monitor the heel lift treatment in children carefully.
- Anterior Lift therapy: affects pelvic rotation – to the same side as the lift.
- Overall Goal – reeducate patient towards an ideal posture.

Treatment of “Short Leg Syndrome”

- OMT – relax & stretch muscles, correct somatic dysfunction, increase mobility
- Exercise – to stretch & tone
- Lift therapy – based on sacral base unleveling, length of time present, amount of compensation

What can correcting a short extremity do?

- Correct many other postural defects by:
 - Shifting body weight
 - Shifting pelvis back to midline
 - Realigning the center of gravity



Summary Slide

- Introduction to “Short Leg Syndrome” and heel lift therapy, with consideration of the potentially involved anatomical areas.
- Practice calculating heel lift height with David Heilig’s formula.

References

- **Required reading: DiGiovanna, EL, Amen, CJ, and Burns, DK, An Osteopathic Approach to Diagnosis and Treatment, Fourth edition, Wolters Kluwer, 2021, Chapter 58 – “Gait and Postural Considerations,”** read from the Heading of “Postural Balance” to the end of the Chapter, LINK:
<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248813979&bookid=2969#248814030>
- **Reference:** Moore, Keith L. and Dalley, Arthur F., Clinically Oriented Anatomy, Fifth Edition, Lippincott, Williams & Wilkins, 2006.
- **Required Reading:** Seffinger, Michael A., D.O., executive editor, **Foundations of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, Fourth Edition, Wolters Kluwer, 2018, Chapter 29: “Postural Considerations in Osteopathic Diagnosis and Treatment”** by Michael L. Kuchera, read from the Section Header of “Triplanar Postural Considerations: Integrated Postural Treatment of Specific Clinical Conditions” to just before the Heading of “Structural/Pathological Considerations: Scoliosis,” LINK:
<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547478>

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Medicine



Questions for Review

Questions for Review

1. "Short Leg Syndrome" is always due to one leg being shorter than the other leg.

- A True
- B False

Questions for Review

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- A True
- ***B False**

Questions for Review

2. "Short Leg Syndrome" is a syndrome, since it has multiple symptoms and findings, such as

- A Sacral base unleveling, cervical spine anterior translation, bilateral tibial torsion
- B Sacral base leveling, innominate flaring, cranial curve side-bending away from the innominate outflaring
- C Sacral base unleveling, compensatory innominate rotation, caudal curve side-bending away from the side of the low sacral base in most cases
- D Sacral base leveling, bilateral femoral anteversion compensation in all cases, medial longitudinal arches decreased in all cases

Questions for Review

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Questions for Review

3. What possible vertebral responses to a postural asymmetry, whether from an actual short leg or from a postural imbalance/somatic dysfunction, can occur?

Questions for Review

4. What could possibly occur to some ligaments over time in cases of "Short Leg Syndrome" or postural asymmetry, where functional compensations become structural?

- A torsion
- B rupture
- C calcification
- D no change

Questions for Review

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Questions for Review

5. Which of the following is matched correctly with regard to counterbalancing muscle pull on the innominates?

- A hamstrings and iliopsoas
- B hamstrings and abdominal muscles
- C spinal extensors and abdominal muscles

Questions for Review

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- ***B hamstrings and abdominal muscles**
- C spinal extensors and abdominal muscles

Questions for Review

6. What information can you gain from observing a person's stance and gait?

Questions for Review

7. To check for a true leg length discrepancy, you can measure from a fixed point to a fixed point. Specifically from the

- A Iliac crest to the medial malleolus
- B Iliac crest to the lateral malleolus
- C ASIS to the medial malleolus
- D ASIS to the lateral malleolus
- E PSIS to the medial malleolus
- F PSIS to the lateral malleolus

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Questions for Review

8. The Hip Drop Test is utilized to assess

- A Hip joint involvement
- B Lumbar involvement
- C Innominate involvement
- D Sacral involvement

Questions for Review

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- ***B Lumbar involvement**
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Questions for Review

9. A heel lift is added to the heel on the side of

- A the elevated iliac crest
- B the lowered iliac crest
- C the elevated sacral base
- D the lowered sacral base
- E the anteverted femoral head

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9. A heel lift is added to the heel on the side of

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- B the lowered iliac crest
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Questions for Review

10. The lumbosacral compensation in "Short Leg Syndrome" or postural asymmetry always involves caudal-curve side bending away from the side of the low sacral base

- A True
- B False

Questions for Review

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- A True
- ***B False**

Questions for Review

11. What is the maximum amount of heel lift that should be placed inside a shoe?

- A $\frac{1}{16}$ inch
- B $\frac{1}{10}$ inch
- C $\frac{1}{8}$ inch
- D $\frac{1}{4}$ inch
- E $\frac{1}{2}$ inch
- F $\frac{3}{4}$ inch

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Questions for Review

12. What is the maximum amount of heel lift between the patient's heel and the floor, that can be used?

- A 1/8 inch
- B 1/4 inch
- C 1/2 inch
- D 3/4 inch
- E 5/8 inch
- F 7/8 inch

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12. What is the maximum amount of heel lift between the patient's heel and the floor, that can be used?

- A 1/8 inch
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Questions for Review

13. In some cases, greater than a 1/2 inch heel lift may be required. Above 1/2 inch, any further lift requirement is added to the whole shoe (at the heel and the anterior foot portion). The placement of the lift material on the outside of the shoe anteriorly ensures that

- A the innominates will rotate in the transverse plane, moving away from the side of the heel lift
- B the innominates will not rotate excessively in the transverse plane, avoiding the innominates going away from the side of the heel lift
- C the innominates will tilt in the coronal plane, towards the side of the heel lift

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